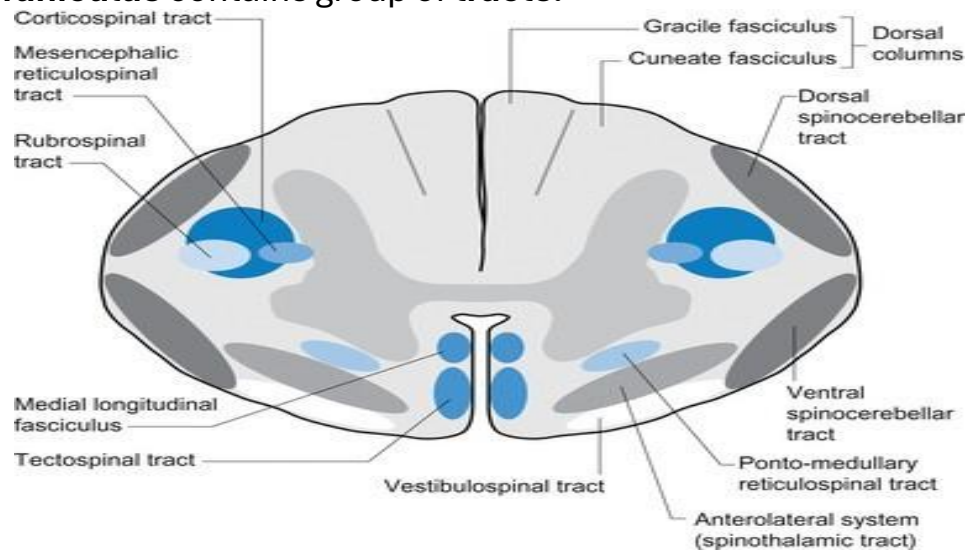


column lies between the entry of the posterior roots and the midline.

- **Each funiculus** contains group of **tracts**.

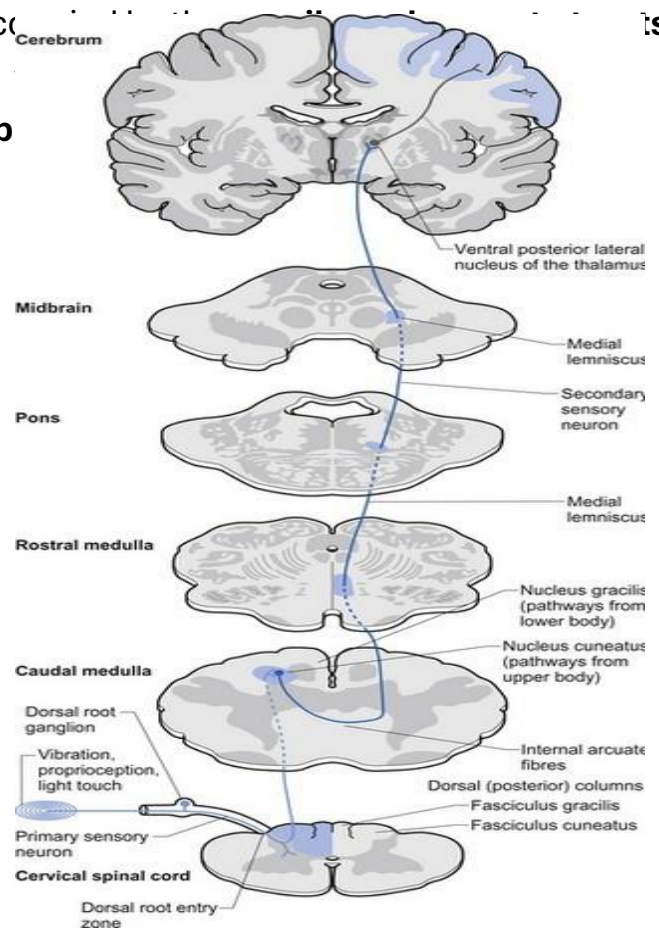


Lecture (12): Spinal cord II

- The **spinal tracts** are divided into:
 - Ascending sensory tracts.
 - Descending motor tracts.
 - Intersegmental short tracts

Ascending sensory tracts:

- The posterior white column is wholly occupied by the **ascending sensory tracts**.
- The main ascending tracts of lateral white column are the **lateral spinothalamic tract, anterior & posterior spino-tectal tracts**.
- The main ascending tracts of anterior white column is the **ventral spinothalamic tract**.
- The sensation from the sensory receptors to the cerebral cortex or subcortical is conducted by series of neurons called pathway.



1- Fasciculus Gracilis (gracile tract) & Fasciculus Cuneatus (cuneate tract):

- **Begin and course:** The central processes from posterior root ganglion pass directly to the posterior white column of the same side and ascend as gracile and cuneate tracts.
- The **gracile tract**: lies close to the midline throughout the length of the spinal cord. It contains long ascending fibers from sacral, lumbar and lower 6 thoracic spinal nerves.
- The **cuneate tract**: is situated lateral to the gracile tract in the upper 6 thoracic and cervical segments of the spinal cord. It contains the long ascending fibers from upper 6 thoracic and all cervical spinal nerves.

➤ Termination:

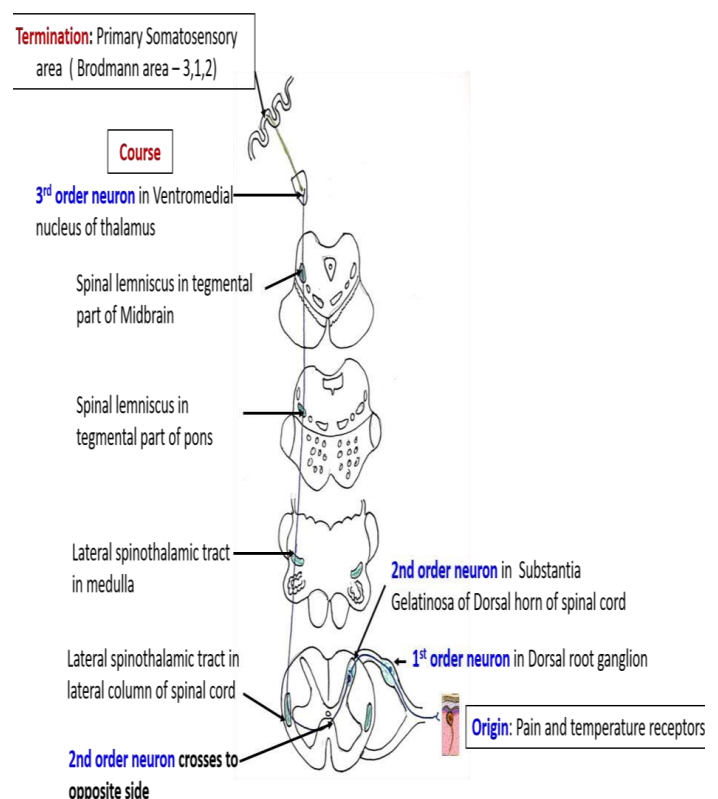
- The **gracile tract** ascends ipsilateral and ends on the gracile nucleus of medulla oblongata.
- The **cuneate tract**: ascends ipsilateral and ends on the cuneate nucleus of medulla oblongata, lateral to cuneate nucleus.
- The **axons of gracile and cuneate nuclei** form the **internal arcuate fibers**, which decussate with the corresponding fibers of the opposite side in the **sensory decussation**.
- The fibers then ascend as the **medial lemniscus**. Its fibers terminate on the ventral posterolateral nucleus of the thalamus. The axons from this nucleus reach the postcentral gyrus (area 3,1,2) of the cerebral cortex.

➤ Functions:

- The **gracile tract** is concerned with fine touch, vibration sense and conscious proprioception from the lower half of the same side of the body.
- The **cuneate tract** conveys these sensations from upper half of the same side of the body.

2- Lateral spino-thalamic tract:

➤ Begin and course:



- The central processes from posterior root ganglion proceed to the tip of the posterior horn and divide into ascending & descending fibers to form posterolateral tract of Lissauer which terminates by synapsing with substantia gelatinosa of Rolandi.
- Axons of SGR cross obliquely to the opposite side in front of the central canal, ascending in the contralateral white column as the lateral spinothalamic tract.

➤ Termination:

- lateral spinothalamic tract ascends through the spinal cord and medulla oblongata, it is now accompanied by the anterior spinothalamic tract, together they form the spinal lemniscus.
- Its fibers terminate by synapsing on the ventral posterolateral nucleus of the thalamus. The axons from this nucleus reach the postcentral gyrus (area 3,1,2) of the cerebral cortex.

➤ Functions:

- It carries pain and temperature sensations from the opposite side of the body.

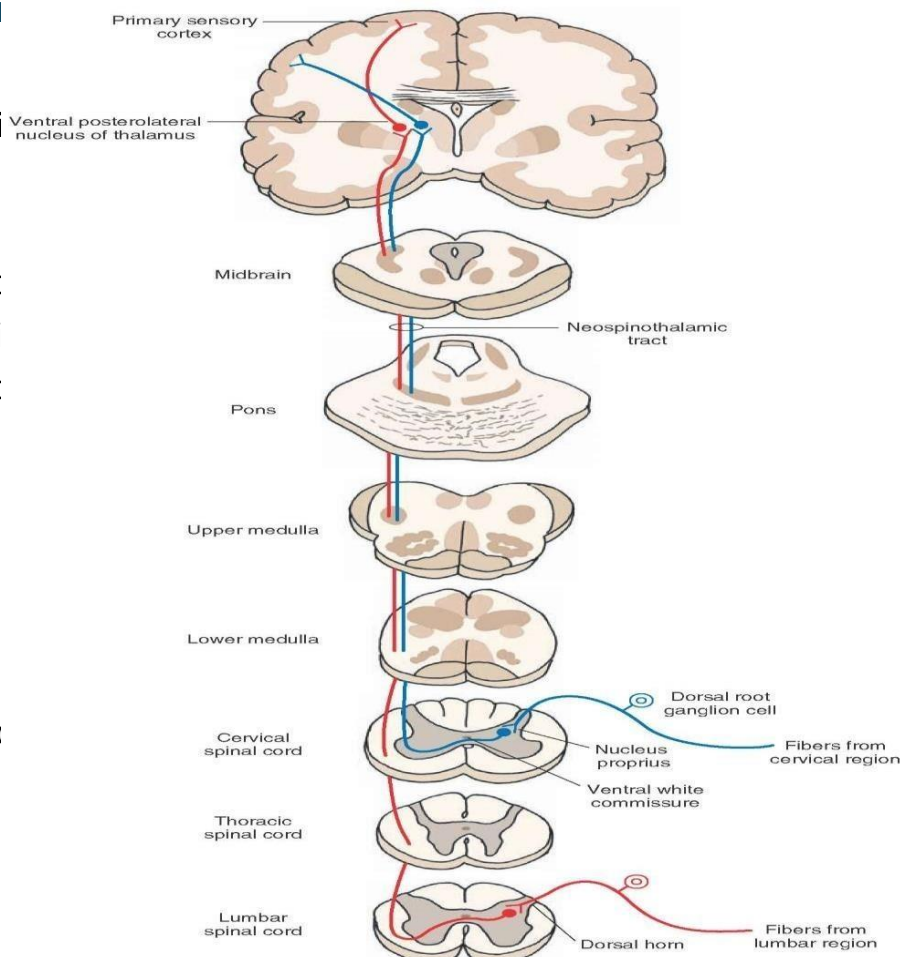
3- Anterior Spinothalamic Tract

➤ Begin and course:

- The axons enter the spinal cord to the tip of posterior horn fibers.
- These fibers terminate in the nucleus proprius.
- Axons of nucleus proprius cross in contralateral anterior white column as anterior spinothalamic tract.

➤ Termination:

- Anterior spinothalamic tract ascends through the spinal cord and medulla oblongata, it is now accompanied by the lateral spinothalamic tract, together they form the spinal lemniscus.



they form the **spinal lemniscus**. Its fibers terminate by synapsing on the ventral posterolateral nucleus of the thalamus. These axons from this nucleus reach the postcentral gyrus (area 3,1,2) of the cerebral cortex.

➤ Functions:

- It carries crude touch sensation from the opposite side of the body.

N.B: These tracts end at subcortical centers for functional integration.

Tract	Posterior spinocerebellar tract	Anterior spinocerebellar tract	Spino-olivary tract	Spino-tectal tract
Begin	It is formed of axons of Clarke's nucleus.	It is formed of axons of spinal border cells.	It is formed of axons of unknown nucleus of PHCs.	It is formed of axons of unknown nucleus of PHCs.
Course	It ascends in Lateral white column ipsilateral.	These axons cross to opposite side in Lateral white column.	It ascends in lateral white column.	It ascends in Lateral white column.
End	Passes through inferior cerebellar peduncle to the Cerebellum	It recrosses through superior cerebellar peduncle to end in Cerebellum	Inferior olivary nucleus of medulla oblongata	Superior colliculus of midbrain
Function	It transmits proprioception from the trunk & lower limbs.	It transmits proprioception from the lower limbs.	It Conveys proprioceptive information from spinal cord to cerebellum via inferior olivary nucleus.	It provides afferent information for Spino-visual reflexes.

Descending motor tracts:

▪ **Pyramidal tract**

- It is composed of **corticospinal tract & corticobulbar tract**.
- Control of skeletal muscle activity and fine (skilled) movements especially those of distal parts of the limbs.
- Originates from cerebral cortex.

Corticospinal tract:

➤ **Beginning:**

Corticospinal tract is formed mainly of axons of **giant pyramidal cells of Betz**:

- One-third of fibers of corticospinal tract originate from **primary motor cortex (area 4)**.
- One-third originate from the **premotor area (area 6)**.
- One-third originate from **primary somatosensory area (area 3, 1, 2)**.

➤ Course:

- It descends through **corona radiata** then in the **posterior limb of internal capsule**.
- It descends through the **middle 3/5 of crus cerebri of midbrain**.
- It descends in the **basis pontis of the pons**.
- It forms the **pyramid** in the medulla oblongata.
- In the lower part of closed medulla: **about 85% of the corticospinal axons** decussate to opposite side forming the **pyramidal decussation (motor decussation)** and continue caudally as the **lateral corticospinal tract**. The rest of the fibers continues uncrossed as the **anterior corticospinal tract**.

➤ Termination:

- The **lateral corticospinal tract** descends in the lateral funiculus of the spinal cord, terminating in the anterior horn cells.
- The **anterior corticospinal tract** descends in the anterior funiculus of the spinal cord then decussates and terminates in the upper thoracic segments.

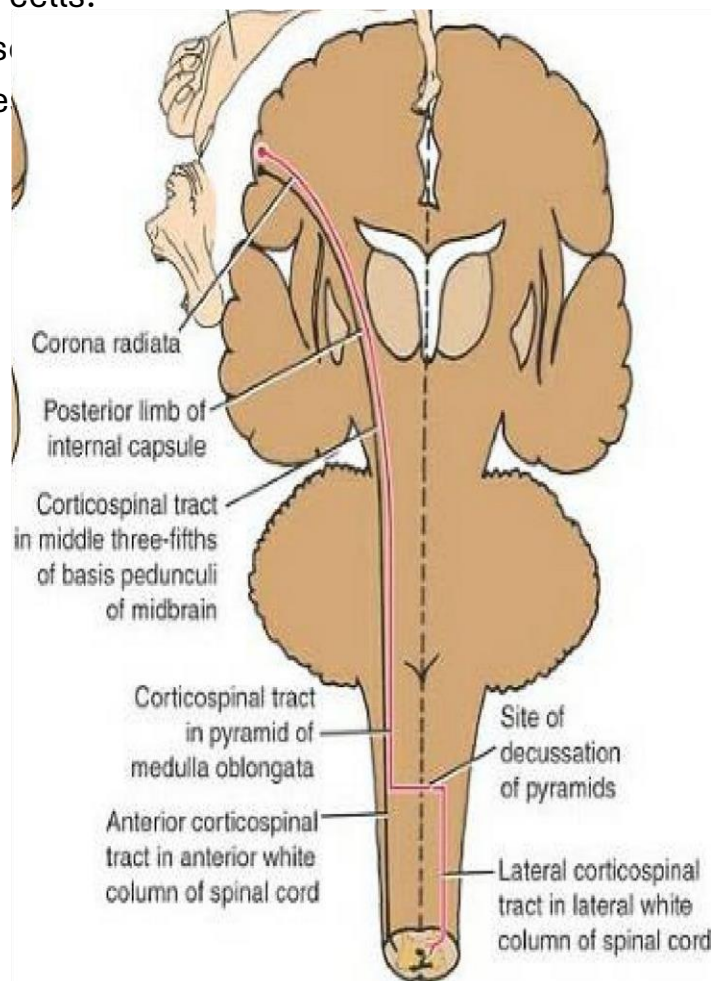
➤ Functions:

- **Lateral corticospinal tract** is concerned with generating voluntary, skilled movements of distal muscles of limbs.
- **Anterior corticospinal tract** controls voluntary movements of trunk muscles.

Lesion to corticospinal tract:

(upper motor neuron lesion):

✓ Above motor decussation:



contralateral hemiplegia.

✓ Below motor decussation:

ipsilateral hemiplegia.

- **55% of fibers of corticospinal tract** terminate at cervical region, **20%** at thoracic and **25%** at lumbosacral level.
- **Upper motor neurons** are giant pyramidal cells of Betz and their axons (pyramidal tract).
- **Lower motor neurons** are AHC and their axons supply skeletal muscles.

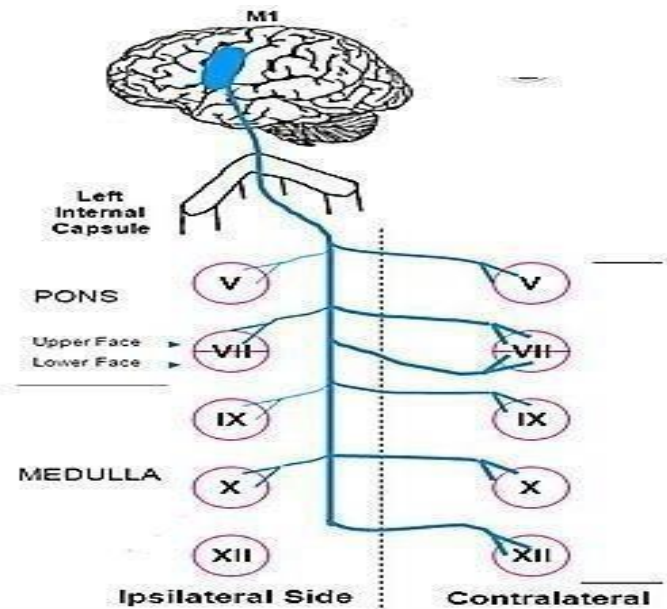
Corticobulbar (corticonuclear) tract:

➤ It is divided into 2 tracts:

1- Medial corticobulbar tract.

2- Lateral corticobulbar tract.

- **Medial corticobulbar tract** arises from frontal eye field (area 8). It ends on nuclei of the 3rd, 4th, 6th cranial nerves supplying extraocular muscles.
- **Lateral corticobulbar tract** arises from motor cortex (area 4). It ends on the motor nuclei of the 5th, 7th, 9th, 10th, 11th, 12th cranial nerves, supplying muscles of head & neck.
- **Fibers of corticobulbar tract** end bilaterally on the motor cranial nerve nuclei except, hypoglossal nucleus and lower ½ of facial nucleus receive only contralateral supply.



Extrapyramidal tracts

- They originate from **subcortical motor centers** in the brainstem, descend and terminate on AHCs of spinal cord.
- They are concerned with **gross and reflex movements**.

1- Tecto-spinal tract:

- **Origin:** Superior colliculus in the midbrain.
- **Functions:** Concerned with coordination of the head and eyes movements in response to visual, auditory or somatic stimuli.

2- Rubro-spinal tract:

- **Origin:** Red nucleus in the midbrain.
- **Course:** It descends through the pons and medulla oblongata to enter the lateral white column of the spinal cord.
- **Functions:** Facilitates the tone of limb flexors and inhibits the activity of the extensors. Assist Crawling in babies.

3- Medial reticulo-spinal tract:

- **Origin:** Reticular nuclei of Pons.
- **Functions:** facilitates tone of extensors.

4- Lateral reticulo-spinal tract:

- **Origin:** Reticular nuclei of medulla oblongata.
- **Functions:** facilitates tone of flexors.

5- Medial vestibulo-spinal tract:

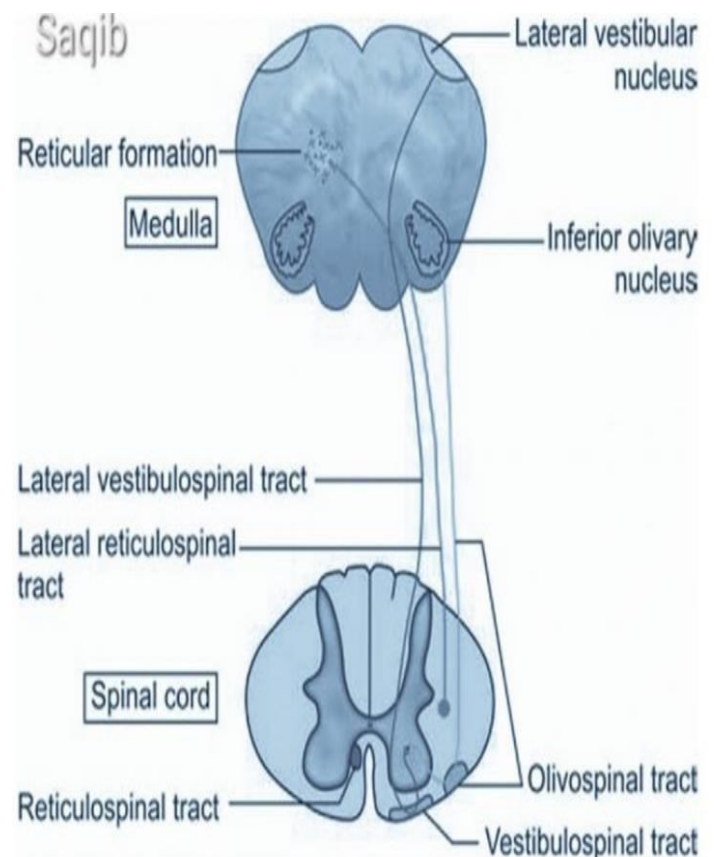
- **Origin:** Medial vestibular nucleus located in medulla.
- **Functions:** Coordination of head, neck & eye movements.

6- Lateral vestibulo-spinal tract:

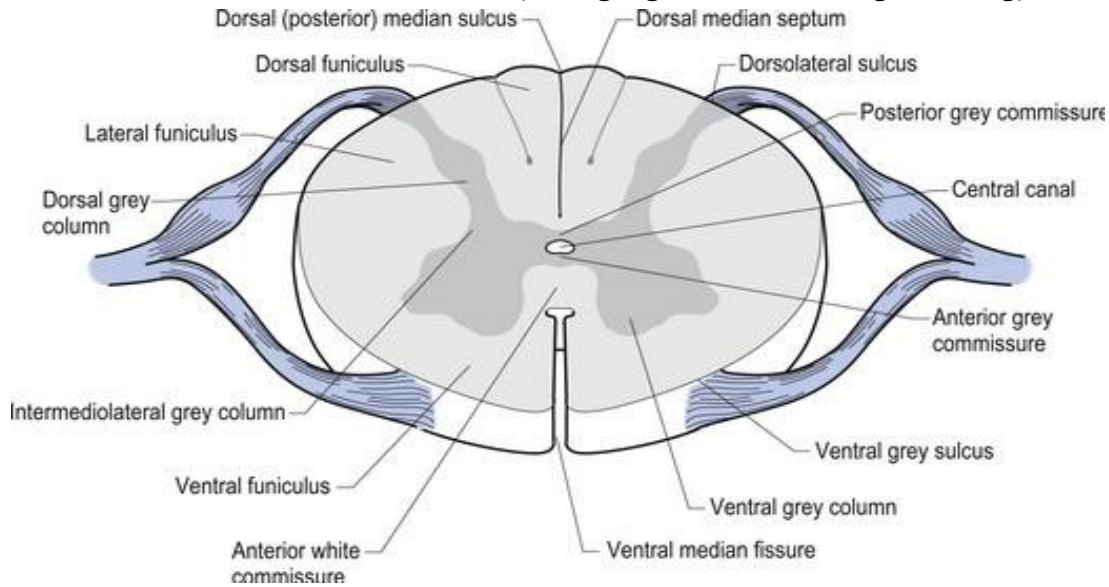
- **Origin:** Lateral vestibular nucleus located in pons.
- **Functions:** Facilitatory to extensors of the back to maintain posture and balance.

7- Oливо-spinal tract:

- **Origin:** Inferior olivary nucleus located in medulla.
- **Course:** descends in lateral white column of spinal cord to terminate on **AHCs**.



- **Functions:** Reflex arm movement (swinging of arms during walking).



Lecture

(12):

Peripheral nervous system

- **The peripheral nervous system (PNS)** is one of two components that make up the nervous system, with the other one is the central nervous system (CNS).
- The **PNS** is composed of:
 - 1- **Cranial nerves.**
 - 2- **Spinal nerves.**
 - 3- **Their associated ganglia outside the CNS.**
- These nerves form the communication network between the central nervous system and various body parts.
- **Functionally**, the **PNS** is subdivided into:
 - 1- **Somatic nervous system.**
 - 2- **Autonomic nervous system.**

Somatic nervous system :

- The somatic nervous system is the voluntary part in controlling skeletal muscles

and processing of somatic sensation.

- The major structures of the PNS are the peripheral receptors, the neuromuscular junction, the peripheral nerves (trunks and plexuses) and their associated sensory ganglia.
- **Peripheral receptors** can be classified by sensory modality as mechanoreceptors, chemoreceptors or photoreceptors. Each receptor is generally specialized to be sensitive to a specific type of stimulus.
- It consists of:
 - 1- **Sensory nerves** carrying afferent nerve fibers, which convey sensation from the body to the central nervous system (CNS).
 - 2- **Motor nerves** carrying efferent nerve fibers, which carry motor commands from the CNS to stimulate muscle contraction.
- **Cranial nerves:**

There are **12 pairs of cranial nerves that** emerge from the cranium. They carry information into and out of the brain.
- **Spinal nerves:**

There are **31 pairs** of spinal nerves: **8 cervical, 12 thoracic, 5 lumbar, 5 sacral** and **1 coccygeal**. They exit from the spinal cord. They carry sensory information into motor commands out of the spinal cord.

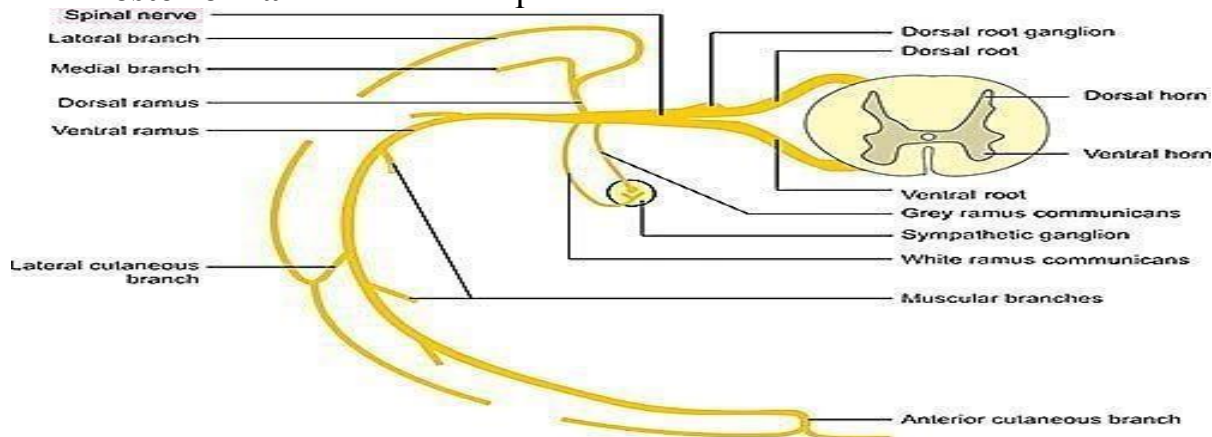
Structure of typical spinal nerve:

- Spinal nerves are formed by the union of dorsal and ventral roots; thus, most **spinal nerves are mixed nerves** containing both sensory and motor fibers. However, the **first cervical spinal nerve** does not have a sensory root, and the **fifth sacral** and **coccygeal spinal nerves** do not have ventral roots.
- **Dorsal roots** transmit sensory information from skin, muscle, and joints to cells in the grey matter of the spinal cord.

- The **dorsal root** contains central processes of a swelling called the dorsal root ganglion that houses the cell bodies of all the sensory neurons that send axons into the spinal cord.
- **Ventral roots** transmit motor and autonomic information to the periphery.
- The **anterior root** is the efferent motor root. It contains:
 - Motor efferent fibers (axons of anterior horn cells) for skeletal muscles.
 - Preganglionic sympathetic fibers in ventral roots of (T1-L3).
 - Preganglionic parasympathetic fibers in ventral roots of S 2,3,4.
- In the **upper cervical region**, the spinal nerve roots are short and run almost horizontally.
- Roots of **the lumbar and sacral nerves** below the level of the termination of the cord (lower border of the first lumbar vertebra in the adult) form a **vertical leash of nerves** around the filum terminale. Together, these lower nerve roots are called the **cauda equina**.
- The spinal nerve roots pass from the spinal cord to the level of their **respective intervertebral foramina**, where they unite to form a **spinal nerve trunk**. Here, the motor and sensory fibers become mixed together; thus, a spinal nerve is made up of a mixture of motor and sensory fibers.
- Spinal nerves divide into **dorsal and ventral rami**. Each ramus contains both sensory and motor fibers.
- **After emerging** from the intervertebral foramen, each spinal nerve immediately divides into a **large anterior ramus** and a **smaller posterior ramus**, each containing both motor and sensory fibers.
- The **posterior ramus passes** posteriorly around the vertebral column to supply the muscles and skin of the back.
- The **anterior ramus** continues anteriorly to supply the muscles and skin over the anterolateral body wall and all the muscles and skin of the limbs.

- The **anterior rami** join one another at the root of the limbs to form complicated nerve plexuses. The **cervical and brachial plexuses** are found at the root of the upper limbs, and **the lumbar and sacral plexuses** are found at the root of the lower limbs.

▪ **Posterior rami** do not form plexuses.

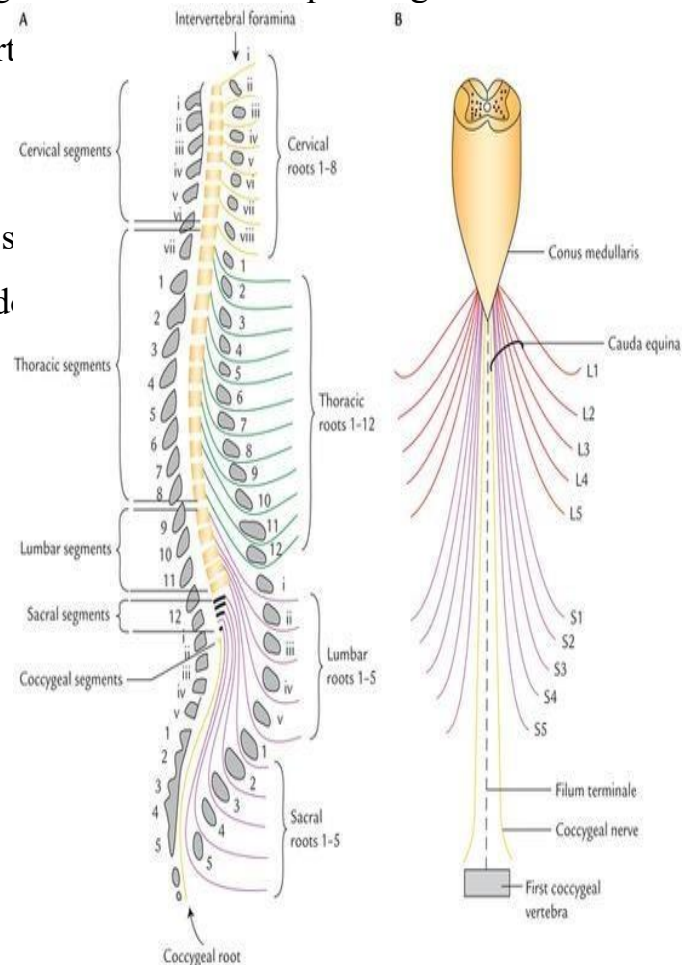


Mode of exit of spinal nerves:

- Upper 7 cervical nerves (C1-C7) emerge **above** their corresponding vertebrae.
- **C8 nerve** emerges between C7 & T1 vert
- **T1-L5 nerves** emerge through the corresponding vertebrae.
- **S1- S4 nerves** are divided inside the s through the anterior sacral foramina & d sacral foramina.
- **S 5 & coccygeal nerves** emerge through the **sacral hiatus**.

Dermatomes:

- Each spinal nerve innervates a specific skin region called a dermatome, and the muscles innervated by a single ventral root are called a myotome.
- Most muscles are innervated by axons

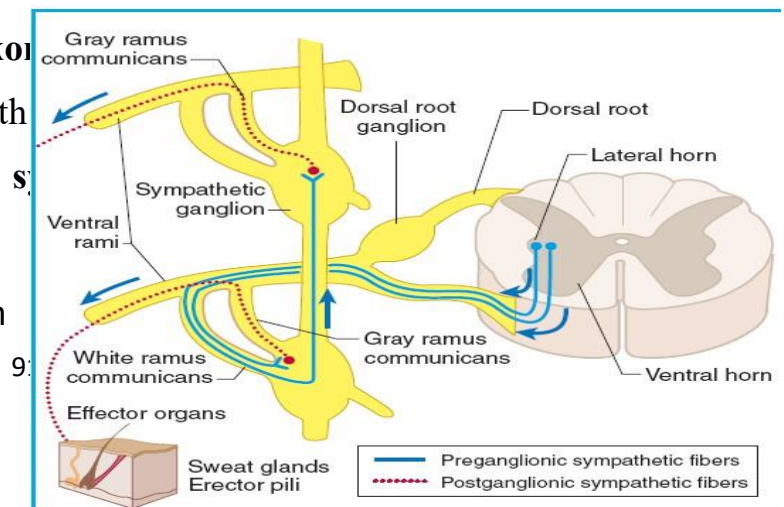


from two spinal segments.

- Muscles sharing a common primary action are supplied by the same segment and oppose muscles by segments in sequence with the former. For example, in elbow movement, spinal cord segments C5 and C6 supply the bicep flexors and C7 and C8 innervate the triceps extensors.

Autonomic nervous system (ANS)

- The autonomic nervous system is the part of the nervous system concerned with the innervation of involuntary structures, such as the **heart, smooth muscle, and glands** within the body.
- It is distributed throughout the central and peripheral nervous systems.
- The autonomic system may be divided into two parts, the **sympathetic** and the **parasympathetic**, and in both parts, there are afferent and efferent nerve fibers.
- The activities of the sympathetic part of the autonomic system prepare the body for an emergency.
- The activities of the parasympathetic part of the autonomic system are aimed at conserving and restoring energy.
- The **first-order autonomic neurons** have their cell bodies located in the CNS, either in the lateral horn of the spinal cord or in the brain stem. They give rise to **myelinated pre-ganglionic axons** that run from the CNS to synapse on second-order neurons in a ganglion, located completely outside the CNS.
- The **second-order autonomic neurons** are located in the **PNS** & their axons are **unmyelinated post-ganglionic axons**.
- **All autonomic nerves** synapse with the autonomic ganglia.
- The ANS has two major divisions: **Sympathetic** and **Parasympathetic**.
- **The sympathetic division:**
- The sympathetic nervous system



supplies **visceral structures**

throughout the entire body.

- It supplies visceral structures associated with the skin (sweat glands, blood vessels, and arrector pili muscles) and deeper visceral structures of the body (blood vessels, smooth muscle in the walls of organs, and various glands).
- The **sympathetic preganglionic cell bodies** are in the **lateral horns** of **T1 - L2 segments** the spinal cord.
- The **preganglionic fibers** leave the spinal cord through the ventral roots to the corresponding spinal nerve. This outflow is referred to as the **thoraco-lumbar outflow**.
- The **myelinated preganglionic fibers** from the ventral ramus to the sympathetic trunk is called a **white ramus communicans**.
- The postganglionic sympathetic fibers leave the sympathetic chain to enter all 31 pairs of spinal ventral rami by means of the 31 pairs of gray rami **communicans**. fibers then project onto their target structures.
- **Ganglia of sympathetic system** lies in two different localities:
 - **Two sympathetic chains:** one on either side of the vertebral column, extending from 1st cervical vertebra above to the front of coccyx below, where the two chains fuse to form **single coccygeal ganglion impar**. There are **23 pairs of ganglia** can be found: **3** in cervical region (superior, middle and inferior cervical ganglions), **12** in thoracic region, **4** in lumbar region, **4** in sacral region, and a **single ganglion impar**.
 - **Prevertebral (Collateral) ganglia or preaortic ganglia:** around large arteries arising from abdominal aorta as **coeliac & mesenteric ganglia**.
- **The parasympathetic division:**
- The parasympathetic portion of the autonomic nervous system is called the **craniosacral outflow** because it has its preganglionic cell bodies in the

brainstem and in the **sacral portion** of the spinal cord.

- Parasympathetic fibers run in **cranial nerves III, VII, IX, and X**.
- The sacral parasympathetic outflow arises from the intermediolateral horn of sacral spinal cord segments **2, 3, 4**, and its fibers are called **pelvic splanchnic nerves**. which supply lower part of the gut, urinary bladder & genitalia.
- Preganglionic parasympathetic fibers are **long** while the postganglionic fibers are **short**.
- Ganglia of parasympathetic system (**terminal ganglia**) are present on or in the wall of the organs which they supply.
- There are **four parasympathetic ganglia** in the head & neck:
 - 1- **Ciliary ganglion**: it lies in the **orbit**, its preganglionic fibers arise from **Edinger-Westphal nucleus** in midbrain, pass with **oculomotor nerve**, its postganglionic fibers supply **sphincter pupillae muscle & ciliary muscle**.
 - 2- **Pterygopalatine ganglion**: it lies in the **pterygopalatine fossa**, its preganglionic fibers arise from **superior salivatory nucleus** in **pons**, pass with **facial nerve**, its postganglionic fibers supply lacrimal, nasal, palatine, pharyngeal glands.
 - 3- **Submandibular ganglion**: it lies in **submandibular region**, its preganglionic fibers arise from **superior salivatory nucleus** in **pons**, pass with **facial nerve**, its postganglionic fibers supply **submandibular & sublingual** salivary glands.
 - 4- **Otic ganglion**: it lies in **infratemporal fossa**, its preganglionic fibers arise from **inferior salivatory nucleus** in **medulla**, pass with glossopharyngeal nerve (CN IX), its postganglionic fibers supply **parotid gland**.
- **Dorsal motor nucleus of vagus** supplies thoracic & abdominal viscera.
- Parasympathetic system supplies **deep viscera only**; does not supply skin or limbs.